

ECON 1001 — Spring 2026 — Midterm 2 Grading Materials

Here all versions of the free response problems, for grading reference.

Version A

22. Bob and Linda run a diner. They spend their time making Burgers and baking Cookies. The table below shows their maximum production of each good per 8-hour shift.

Worker	Burgers per shift	Cookies per shift
Bob	20	60
Linda	40	80

- (a) Who has the absolute advantage in making Burgers? Briefly explain why. Box your answer (for this and each subsequent question).
 - (b) Compute Bob's opportunity cost of making 1 Burger. Show your work.
 - (c) Who has the comparative advantage in baking Cookies? Briefly explain why.
 - (d) Suppose Bob and Linda decide to specialize and trade with one another. What is the range of prices (in terms of cookies) they would agree upon to trade 1 Burger?
23. The table below shows the Marginal Cost (MC) of producing widgets for three different firms. Widgets are sold in a competitive market for **\$60 each**. Initially, the government gives each firm exactly **2 permits**. One permit is required to produce one widget. (Assume firms cannot produce partial widgets.)

Widget Number	Firm A's MC	Firm B's MC	Firm C's MC
1st Widget	\$10	\$15	\$20
2nd Widget	\$25	\$30	\$50
3rd Widget	\$40	\$45	\$70

- (a) Initially, each firm uses its 2 permits to produce 2 widgets. What is the combined total cost of production for these 6 widgets?
- (b) Suppose no permits have been traded yet. What is Firm A's **maximum** willingness-to-pay (WTP) for a 3rd permit, and what is Firm C's **minimum** willingness-to-accept (WTA) to sell one of its 2 permits?
- (c) Assume transaction costs are zero and firms are allowed to trade permits. After all mutually beneficial trades have occurred, what is the new total cost of production for the 6 widgets?

Version B

22. Bob and Linda run a diner. They spend their time making Burgers and baking Cookies. The table below shows their maximum production of each good per 8-hour shift.

Worker	Burgers per shift	Cookies per shift
Bob	10	40
Linda	30	90

- (a) Who has the absolute advantage in making Burgers? Briefly explain why. Box your answer (for this and each subsequent question).
 - (b) Compute Bob's opportunity cost of making 1 Burger. Show your work.
 - (c) Who has the comparative advantage in baking Cookies? Briefly explain why.
 - (d) Suppose Bob and Linda decide to specialize and trade with one another. What is the range of prices (in terms of cookies) they would agree upon to trade 1 Burger?
23. The table below shows the Marginal Cost (MC) of producing widgets for three different firms. Widgets are sold in a competitive market for **\$60 each**. Initially, the government gives each firm exactly **2 permits**. One permit is required to produce one widget. (Assume firms cannot produce partial widgets.)

Widget Number	Firm A's MC	Firm B's MC	Firm C's MC
1st Widget	\$12	\$16	\$22
2nd Widget	\$20	\$32	\$48
3rd Widget	\$38	\$44	\$75

- (a) Initially, each firm uses its 2 permits to produce 2 widgets. What is the combined total cost of production for these 6 widgets?
- (b) Suppose no permits have been traded yet. What is Firm A's **maximum** willingness-to-pay (WTP) for a 3rd permit, and what is Firm C's **minimum** willingness-to-accept (WTA) to sell one of its 2 permits?
- (c) Assume transaction costs are zero and firms are allowed to trade permits. After all mutually beneficial trades have occurred, what is the new total cost of production for the 6 widgets?

Version C

22. Bob and Linda run a diner. They spend their time making Burgers and baking Cookies. The table below shows their maximum production of each good per 8-hour shift.

Worker	Burgers per shift	Cookies per shift
Bob	25	50
Linda	50	150

- (a) Who has the absolute advantage in making Burgers? Briefly explain why. Box your answer (for this and each subsequent question).
 - (b) Compute Bob's opportunity cost of making 1 Burger. Show your work.
 - (c) Who has the comparative advantage in baking Cookies? Briefly explain why.
 - (d) Suppose Bob and Linda decide to specialize and trade with one another. What is the range of prices (in terms of cookies) they would agree upon to trade 1 Burger?
23. The table below shows the Marginal Cost (MC) of producing widgets for three different firms. Widgets are sold in a competitive market for **\$70 each**. Initially, the government gives each firm exactly **2 permits**. One permit is required to produce one widget. (Assume firms cannot produce partial widgets.)

Widget Number	Firm A's MC	Firm B's MC	Firm C's MC
1st Widget	\$15	\$20	\$25
2nd Widget	\$30	\$35	\$60
3rd Widget	\$45	\$50	\$80

- (a) Initially, each firm uses its 2 permits to produce 2 widgets. What is the combined total cost of production for these 6 widgets?
- (b) Suppose no permits have been traded yet. What is Firm A's **maximum** willingness-to-pay (WTP) for a 3rd permit, and what is Firm C's **minimum** willingness-to-accept (WTA) to sell one of its 2 permits?
- (c) Assume transaction costs are zero and firms are allowed to trade permits. After all mutually beneficial trades have occurred, what is the new total cost of production for the 6 widgets?

Version D

22. Bob and Linda run a diner. They spend their time making Burgers and baking Cookies. The table below shows their maximum production of each good per 8-hour shift.

Worker	Burgers per shift	Cookies per shift
Bob	15	60
Linda	20	40

- (a) Who has the absolute advantage in making Burgers? Briefly explain why. Box your answer (for this and each subsequent question).
 - (b) Compute Bob's opportunity cost of making 1 Burger. Show your work.
 - (c) Who has the comparative advantage in baking Cookies? Briefly explain why.
 - (d) Suppose Bob and Linda decide to specialize and trade with one another. What is the range of prices (in terms of cookies) they would agree upon to trade 1 Burger?
23. The table below shows the Marginal Cost (MC) of producing widgets for three different firms. Widgets are sold in a competitive market for **\$80 each**. Initially, the government gives each firm exactly **2 permits**. One permit is required to produce one widget. (Assume firms cannot produce partial widgets.)

Widget Number	Firm A's MC	Firm B's MC	Firm C's MC
1st Widget	\$20	\$25	\$30
2nd Widget	\$30	\$35	\$65
3rd Widget	\$40	\$55	\$85

- (a) Initially, each firm uses its 2 permits to produce 2 widgets. What is the combined total cost of production for these 6 widgets?
- (b) Suppose no permits have been traded yet. What is Firm A's **maximum** willingness-to-pay (WTP) for a 3rd permit, and what is Firm C's **minimum** willingness-to-accept (WTA) to sell one of its 2 permits?
- (c) Assume transaction costs are zero and firms are allowed to trade permits. After all mutually beneficial trades have occurred, what is the new total cost of production for the 6 widgets?

Version E

22. Bob and Linda run a diner. They spend their time making Burgers and baking Cookies. The table below shows their maximum production of each good per 8-hour shift.

Worker	Burgers per shift	Cookies per shift
Bob	30	90
Linda	60	120

- (a) Who has the absolute advantage in making Burgers? Briefly explain why. Box your answer (for this and each subsequent question).
 - (b) Compute Bob's opportunity cost of making 1 Burger. Show your work.
 - (c) Who has the comparative advantage in baking Cookies? Briefly explain why.
 - (d) Suppose Bob and Linda decide to specialize and trade with one another. What is the range of prices (in terms of cookies) they would agree upon to trade 1 Burger?
23. The table below shows the Marginal Cost (MC) of producing widgets for three different firms. Widgets are sold in a competitive market for **\$50 each**. Initially, the government gives each firm exactly **2 permits**. One permit is required to produce one widget. (Assume firms cannot produce partial widgets.)

Widget Number	Firm A's MC	Firm B's MC	Firm C's MC
1st Widget	\$5	\$10	\$15
2nd Widget	\$15	\$20	\$40
3rd Widget	\$25	\$35	\$60

- (a) Initially, each firm uses its 2 permits to produce 2 widgets. What is the combined total cost of production for these 6 widgets?
- (b) Suppose no permits have been traded yet. What is Firm A's **maximum** willingness-to-pay (WTP) for a 3rd permit, and what is Firm C's **minimum** willingness-to-accept (WTA) to sell one of its 2 permits?
- (c) Assume transaction costs are zero and firms are allowed to trade permits. After all mutually beneficial trades have occurred, what is the new total cost of production for the 6 widgets?